

Why Is This Hard? — Metric Ambiguity

Key insight: Two cameras with different focal lengths can produce *identical* images of a scene at different depths.

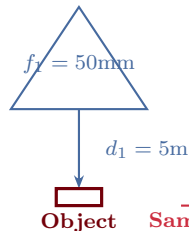
The apparent depth scales with focal length:

$$d_a = \hat{S} \cdot \frac{\hat{f}}{\hat{S}'}$$

where $\hat{S}$ is real-world size, $\hat{f}$ is focal length, and $\hat{S}'$ is projected size.

A network trained on mixed datasets with different cameras sees **conflicting depth labels** for visually identical patches.

Camera 1: f_1



Camera 2: f_2

